

JEE ADVANCED - 3 | PAPER - 2

JEE 2022

Date	Maximum Marks	Duration
1st August, 2021	198	3 Hours

PAPER SCHEME

- The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **three** sections (Section 1, Section 2 & Section 3).
- Section 1** contains **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE**.

Section 2 contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

Section 3 contains **6 Numerical Value Type Questions**. The answer to each of the question is a numerical value. For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)**
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION – 1 (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE**.
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct integer is entered;

Zero Marks : 0 If the question is unanswered;

Negative Marks: –1 In all other cases.

SECTION - 2 (Maximum Marks: 24)

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If only (all) the correct option(s) is(are) chosen

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen and both of which are correct

Partial Marks: +1 If two or more options are correct but ONLY one option is chosen, and it is a correct option

Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)

Negative Marks: –2 In all other cases.

SECTION – 3 (Maximum Marks: 24)

- This section has **SIX (06)** Questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)**
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered;

Zero Marks: 0 In all other cases.

SUBJECT I : PHYSICS

66 MARKS

SECTION 1

SINGLE DIGIT INTEGER

This section consists of **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE**.

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of **SIX (06)** Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has **SIX (06)** Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to **TWO** decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25)

SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY**66 MARKS****SECTION 1****SINGLE DIGIT INTEGER**

This section consists of **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE.**

- In the reaction $X + Y + Z \longrightarrow \text{Products}$.
The time taken for 75% consumption of Y is 1.5 times of that required for its 50% consumption. When simultaneously concentration of both X and Z are doubled, rate of reaction becomes four times. The time taken for 50% consumption of Z becomes half if initial amount of Z is doubled. Overall order of reaction is:
- Consider following nuclear reaction.
I. ${}_{92}^{238}\text{U} \longrightarrow {}_{84}^{218}\text{Po}$ **II.** ${}_{84}^{218}\text{Po} \longrightarrow {}_{83}^{214}\text{Bi}$ **III.** ${}_{83}^{214}\text{Bi} \longrightarrow {}_{82}^{206}\text{Pb}$
 Number of beta particles emitted in reactions I, II and III are x, y and z respectively. Then value of $(x + y - z)$ is :
- Compound "X" have its observed weight in a magnetic balance more than its actual weight determined by using a non magnetic balance. Number of compounds from the following list which can be compound "X" are :
 MnO Fe_3O_4 CrO_2 Mn NaCl ZnS
- Out of given crystal system, the number of crystal systems in which atleast one of the edge length is shorter or longer than the other edge lengths and one of the interaxial angle is necessarily 90° are :
 Cubic Tetragonal Orthorhombic Monoclinic Hexagonal Rhombohedral Triclinic
- To create effectively one Schottky defect per unit cell of NaCl the minimum number of Na^+ ions to be removed is x and minimum number of Cl^- ions to be removed is y then $(x + y)$ is equal to :
- pH of 0.1 M HA solution is 1.70 and pH of 0.2 M HB solution is 1.00 at 25°C . Here HA and HB are weak monobasic acids. Value of $10 \frac{[\pi_{\text{HA}}]}{[\pi_{\text{HB}}]}$ is : [Take $\log_{10} 2 = 0.30$]
 [Here π_{HA} and π_{HB} are osmotic pressures of the solution of HA and HB respectively]

SPACE FOR ROUGH WORK

SECTION 2**MULTIPLE CORRECT ANSWERS TYPE**

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Select the correct order/s :
- (A) Osmotic pressure : $0.1 \text{ M Al}_2(\text{SO}_4)_3 < 0.1 \text{ M K}_3[\text{Fe}(\text{CN})_6]$
- (B) Vapour pressure : 1 molal NaCl in water $<$ 1 molal glucose in water
- (C) Boiling point : 1 molal aqueous NaCl $<$ 1 molal aqueous CaCl_2
- (D) Freezing point : $0.1 \text{ M } [\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3 < 0.1 \text{ M } [\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$
8. Zinc sulphide can crystallize in two type of crystalline structures, zinc blende and wurtzite. If sulphide form cubic close packed lattice in zinc blende and hexagonal close packed lattice in wurtzite and alternate tetrahedral voids are occupied by zinc ions in both then identify incorrect statement(s).
- (A) Coordination number of $\text{Zn}^{2+} = 4$ in both type of lattices
- (B) Zn^{2+} ions are shared by the adjacent unit cells in both type of lattices
- (C) Number of ZnS formula units in each unit cell are equal to 4 and 6 for zinc blende and wurtzite respectively
- (D) 50% of total voids (T.V. + O.V.) are occupied in both lattices
9. Which of the following pairs of liquids will show given properties ?
- | | | |
|--|---|---|
| I. $\Delta H_{\text{mix}} < 0$ | II. $\Delta S_{\text{mix}} > 0$ | III. $\Delta V_{\text{mix}} < 0$ |
| (A) $\text{H}_2\text{SO}_4 + \text{H}_2\text{O}$ | (B) $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_3 + \text{CHCl}_3$ | |
| (C) $\text{HNO}_3 + \text{H}_2\text{O}$ | (D) $\text{CHCl}_3 + \text{CCl}_4$ | |
10. Which of the following statement is/are TRUE regarding collision theory for bimolecular gaseous reaction?
- (A) Rate of a chemical reaction increases with increase in collision frequency between the molecules
- (B) Rate of chemical reaction is independent of change in temperature of the reaction
- (C) Observed frequency factor is greater than the value obtained theoretically from the Arrhenius equation if the orientation factor $P > 1$
- (D) Rate of a chemical reaction increases with the Boltzmann factor $[\exp(-E_A / RT)]$
11. Select correct statement/s from the following :
- (A) Nuclei having $\frac{N}{P} > 1$ undergo alpha emission which further increases the $\frac{N}{P}$ ratio
- (B) K-capture by a parent nuclei gives its isobar
- (C) Neutron emission by a parent nuclei gives its isotope
- (D) Positron emission is a feature of a nuclei with high $\frac{P}{N}$ ratio

12. Select correct statement/s regarding different unit cell arrangements of identical spheres :
- (A) In a 3D HCP unit cell, three spheres are contributing their 100% volume within an unit cell
- (B) In a 3D HCP unit cell, stacking of layers is ABABA..... type and distance between two identical layers is equal to height of the hexagonal unit cell
- (C) Packing fraction of a 2D hexagonal unit cell is higher than the 3D HCP unit cell
- (D) Coordination number of a sphere within its layer, in a 3D HCP unit cell is 6

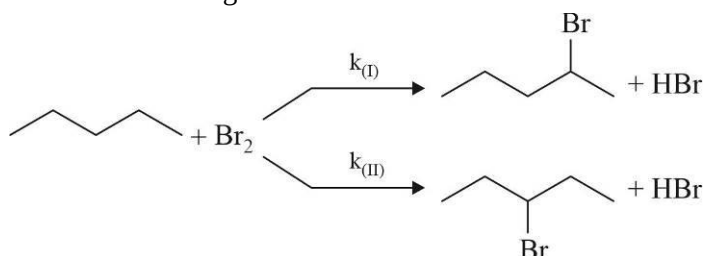
SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

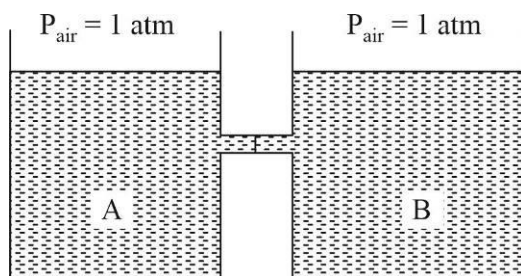
13. Consider following chemical reactions.



If $k_{\text{(I)}} = 8 \times 10^{-2} \text{ min}^{-1}$, $k_{\text{(II)}} = 2 \times 10^{-2} \text{ min}^{-1}$ then percentage of products with a chiral carbon atom will be:

14. A truncated octahedron has x faces, y vertices and z edges. Find the value of $x + y + z$.
15. 2 moles of a liquid $A[P_A^\circ = 100 \text{ torr}]$ mixed with 4 moles of liquid $B[P_B^\circ = 150 \text{ torr}]$. The first distillate obtained on fractional distillation of above mixture was then used as a solvent to prepare 2 molal solution of a non electrolyte and non volatile solute. Vapour pressure of the resulting solution (in torr) is : [Take molar masses (g / mol) : $M_A^\circ = 100$, $M_B^\circ = 80$]
16. In an ionic compound A_2B , B^{2-} ions are forming a F.C.C. lattice while A^{1+} ions are occupying all tetrahedral holes. A new compound "X" is created by doing following sequential structural changes in the compound A_2B . (Neglect charge balance)
- Remove anions from the sites other than the corners
 - Place cation C^{2+} at the sites vacated in the above step
 - Remove A^{1+} ions alternatively from 50% tetrahedral holes
 - Place D^{-1} ions at the vacated tetrahedral holes
- The value of $\left(\frac{\text{number of cations}}{\text{number of anions}} \right)$ in an unit cell of the final compound X is :

17. Two different solutions A and B are separated using a semi-permeable membrane as shown in the figure. Solution A is 0.2 molar $K_4[Fe(CN)_6]$ while solution B is 0.1 molar K_2SO_4 . How much pressure (in atm) must be exerted over surface of one of the solution to prevent any change in their volume due to osmotic flow of water at 300 Kelvin ? [Take $R = 0.0821 \text{ atm} \cdot \text{L} / \text{mol} \cdot \text{K}$]



18. Rate of chemical reaction becomes 4 times when temperature is increased from 27°C to 47°C . Magnitude of slope of the graph constructed between $\log_{10} k$ (y-axis) and $\frac{1}{T}$ (x-axis) is : [Take $\log_{10} 2 = 0.30$]

SPACE FOR ROUGH WORK

SUBJECT III : MATHEMATICS**66 MARKS****SECTION 1****SINGLE DIGIT INTEGER**

This section consists of **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE.**

- The value of $(b+c)$ for which the identity $f(x+1) - f(x) = 8x+3$ is satisfied, where $f(x) = bx^2 + cx + d$, are:
- Let $f(x) = (x-1)^2 \sin \frac{1}{x-1} - |x|$, $x \neq 1$
 $= -1$, $x = 1$
 be a real-valued function. The number of integral points where $f(x)$ is not differentiable is _____.
- $\int \frac{x^2+1}{\sqrt[3]{(x^3+3x+6)}} dx = \frac{1}{a} (x^3+3x+6)^{b/c} + k$, $a, b, c \in N$, where b & c are co-prime. The value of $a+b+c$ is _____.
- If $\int \frac{(x^2-1)}{(x^4+3x^2+1) \tan^{-1}\left(\frac{x^2+1}{x}\right)} dx = k \log \left| \tan^{-1} \frac{x^2+1}{x} \right| + c$, then k is equal to:
- $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx = K_1 \sec x + K_2 \operatorname{cosec} x + C$, then the value of $K_1 + K_2$ is _____.
- The tangents to the curves $y^3 - x^2y + 5y - 2x = 0$ and $x^4 - x^3y^2 + 5x + 2y = 0$ at the origin intersect at an angle θ equal to $\frac{\pi}{k}$. The value k is _____.

SPACE FOR ROUGH WORK

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONE OR MORE THAN ONE CHOICE is correct.

7. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals $-\frac{1}{(\sec x + \tan x)^a} \{b + c(\sec x + \tan x)^2\} + K$: (a, b, c are rationals in lowest form & arbitrary constant K)

- (A) value of a is $\frac{11}{2}$ (B) value of b is $\frac{1}{7}$
 (C) value of c is $-\frac{1}{7}$ (D) value of c is $\frac{1}{7}$

8. If $f : [0, \infty) \rightarrow [0, \infty)$ and $f(x) = \frac{x}{1+x}$, then f is:

- (A) One-one (B) onto (C) many one (D) into

9. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = (x-1)(x-2)(x-3)$ is:

- (A) onto (B) One-one (C) into (D) many one

10. $f(x)$ is defined as follows:

$$f(x) = \begin{cases} (\sin x + \cos x)^{\cos ex}, & -\frac{\pi}{2} < x < 0 \\ a, & x = 0 \\ \frac{\frac{1}{e^x} + \frac{2}{e^x} + \frac{3}{e^{|x|}}}{\frac{2}{ae^x} + \frac{3}{be^{|x|}}}, & 0 < x < \frac{\pi}{2} \end{cases}$$

If $f(x)$ is continuous at $x=0$, choose the correct options.

- (A) The function cannot be continuous at $x=0$ as $LHL \neq RHL$
 (B) The value of a is greater than 2
 (C) The value of b is greater than 2
 (D) The product ab is less than 2

11. The co-ordinates of the point(s) on the graph of the function, $f(x) = \frac{x^3}{3} - \frac{5x^2}{2} + 7x - 4$ where the tangent drawn cut off intercepts from the co-ordinate axes which are equal in magnitude but opposite in sign, is:

- (A) $\left(2, \frac{8}{3}\right)$ (B) $\left(3, \frac{7}{2}\right)$ (C) $\left(1, \frac{5}{6}\right)$ (D) None of these

12. If $f(x) = a^{\{a^{|x|} \operatorname{sgn} x\}}$; $g(x) = a^{[a^{|x|} \operatorname{sgn} x]}$ for $a > 0, a \neq 1$ and $x \in \mathbb{R}$, where $\{ \}$ and $[\]$ denote the fractional part and integral part functions respectively, then which of the following statements can hold good for the function $h(x)$, where $(\ln a)h(x) = (\ln f(x) + \ln g(x))$.
- (A) 'h' is even and increasing if $a > 1$ (B) 'h' is odd and decreasing if $a \in (0, 1)$
 (C) 'h' is even and decreasing if $a \in (0, 1)$ (D) 'h' is odd and increasing if $a > 1$

SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

13. If $f(x) = x(\sqrt{x} - \sqrt{x+1})$, then the value of $f'(0)$ is _____.
14. If $T_r = \frac{1^2 + 2^2 + 3^2 + \dots + r^2}{1^3 + 2^3 + 3^3 + \dots + r^3}$ and $S_n = \sum_{r=1}^n (-1)^r \cdot T_r$, Then $\lim_{n \rightarrow \infty} S_n$ equals _____.
15. If $e^{f(x)} = \frac{10+x}{10-x}$, $x \in (-10, 10)$ and $f(x) = kf\left(\frac{200x}{100+x^2}\right)$, then $k =$ _____.
16. $\lim_{x \rightarrow -\infty} (\sqrt{x^2 - x + 1} + ax - b) = 0$, then the value of ab is _____.
17. Let $f(1) = -2$ and $f'(x) \geq 4.2$ for $1 \leq x \leq 6$. The smallest possible value of $f(6)$ is: _____.
18. $\int \left(\frac{2a+x}{a+x} \right) \sqrt{\frac{a-x}{a+x}} dx = k\sqrt{a^2 - x^2} + la\sqrt{\frac{a-x}{a+x}} + c$, where c is arbitrary constant. The value of kl is _____.

End of JEE Advanced-3 | PAPER-2 | JEE-2022